

Umesh Gupta Pedamallu

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Profile

Enthusiastic and detail-oriented Computer Science graduate with a strong interest in software development, cloud technologies, and problem-solving. Experienced in building full-stack applications, developing AI-powered tools, and working with databases. Passionate about learning new technologies, collaborating in team environments, and contributing to impactful projects through hands-on experience.

Education

Bachelor of Technology in Computer Science and Engineering 2022 – 2026
Honors in Full Stack Development CGPA: 9.32 / 10.0
Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada

Intermediate – Mathematics, Physics, Chemistry 2020 – 2022
Kakaraparti Bhavanarayana Junior College Score: 98.4%

Technical Skills

Programming Languages:	Java, Python, JavaScript
Frameworks & Web Technologies:	React.js, Redux, Tanstack, Node.js, Express.js, Springboot, REST APIs, Docker
Cloud & Platforms:	AWS (EC2, S3, IAM, VPC, CloudWatch)
Databases:	MySQL, MongoDB
AI / ML:	Machine Learning, AI Assisted Coding (Github Copilot, Claude, Gemini)
Developer Tools:	Git, GitHub, Postman, IntelliJ IDE, VS Code, Google Colab
Software Engineering:	OOP, Data Structures & Algorithms, Problem Solving, Agile/Scrum

Professional Experience

Java Full Stack Virtual Intern February 2026 – April 2026
Infosys Springboard Remote

- Engineered a two-role digital agriculture marketplace enabling farmers to manage product listings, pricing, inventory, and sales, while allowing retailers to search produce, place or modify orders, complete transactions, and submit feedback.
- Architected a four-service Spring Boot ecosystem using Config Server, Eureka, and API Gateway for centralized configuration, service discovery, and load-balanced routing.
- Developed five RESTful user-management APIs using Spring Boot, JPA/Hibernate, and MySQL with DTOs, validation, role-based access, and exception handling, while improving maintainability through externalized configuration.

Web Developer & Cloud Integration Intern November 2025 – March 2026
SkillDzire Technologies Pvt. Ltd Remote

- Developed and deployed 3 full-stack applications, improving performance and responsiveness by 25%.
- Integrated cloud services (AWS/Cloud) for hosting, storage, and deployment, reducing manual effort by 40%.
- Designed 10 RESTful APIs with authentication and database integration, reducing data retrieval time by 30%.
- Improved scalability using modular architecture and clean code, reducing bug resolution time by 20%.

Technical Projects

Sepsis Prediction Using Multi-Scale Temporal Convolutional Transformer (MSTCT) ([GitHub Repository](#))
Python, PyTorch, Deep Learning, Time-Series

- Designed and trained a hybrid **Multi-Scale TCN + Transformer** architecture on 40,336 ICU patients (1.5M rows) from the PhysioNet Sepsis dataset for early sepsis onset prediction with an AUROC of 0.882.
- Engineered missingness indicators, time-since-observed features, and 6-hour rolling statistics to handle sparse clinical time-series data, expanding the feature set from 40 to 94 features.
- Implemented a sliding window pipeline (48-hour windows, stride 12) with weighted random sampling and BCEWithLogitsLoss positive-class weighting to address severe class imbalance (5% sepsis prevalence).

LuxeStay – Hotel Booking Platform ([GitHub](#)) *Spring Boot, React, PostgreSQL, Razorpay*

- Engineered a production-grade booking platform for HCL Hackathon, with a layered Spring Boot backend delivering RESTful APIs for search, reservations, payments, and admin operations.

- Secured all auth endpoints with stateless JWT + RBAC and applied Resilience4j rate limiting (10 req/s), eliminating 100% of brute-force attack vectors across login, signup, and password-reset flows.
- Built a concurrency-safe reservation engine using pessimistic/optimistic DB locking and a 60-second scheduled expiry job, eliminating double-booking under high concurrent load.
- Tuned performance with Spring Cache and read-only transactions across 6 heavy-read services; capped paginated endpoints at 50 records with sort-field whitelisting, reducing unnecessary DB round-trips.
- Integrated Razorpay with server-side signature verification and 3 payment modes inside atomic transactions; offloaded email confirmations to an async thread pool (5 threads, 100-task queue), keeping p95 payment latency under 200ms.

TravelLoop – Full-Stack Travel Planning Platform ([GitHub](#))

React 19, Express 5, PostgreSQL, Prisma, TypeScript

- Engineered a full-stack travel planning platform for Odoo Hackathon'26 with 6 core modules — multi-stop itinerary builder, curated city guides, activity assignment, smart checklists, trip notes, and a community inspiration hub.
- Implemented Role-Based Access Control (RBAC) with secure authentication, distinct permission layers for users and admins, and a centralized admin dashboard for platform-wide management.
- Built a type-safe data layer using Prisma ORM + PostgreSQL with auto-generated React Query hooks and TypeScript interfaces, reducing frontend integration boilerplate by 70% and eliminating runtime type errors across codebase.
- Developed a community-driven inspiration hub enabling users to publish itineraries, browse and clone trips from other travellers, and discover curated city guides, driving engagement through user-generated travel content.

Draftory ([GitHub Repository](#))

React, Express.js, MongoDB, JWT

- Architected a secure MERN-based content publishing platform with JWT authentication, RESTful APIs, server-side validation, and hardened HTTP security headers.
- Implemented multi-tier rate limiting (5 attempts / 15 minutes) to prevent brute-force attacks and introduced TTL-based in-memory caching, reducing repeated feed retrieval latency by 80%.
- Designed efficient paginated content delivery for blog feeds, improving average blog load time by 60%.

Achievements & Recognition

- Awarded the Reliance Foundation Scholarship (2022–2026) and the LIC Golden Jubilee Foundation Scholarship (2022–2026) in recognition of academic excellence and outstanding performance.
- Led team to top 3 position in SIH Internal Hackathon 2024.
- Solved 1000+ algorithmic problems and participated in 60 coding contests in various coding platforms, with a maximum LeetCode rating of 1939.
- Winner of two consecutive coding events organized by the ACM chapter, VRSEC.
- Selected for the Preliminary Stage of Summer of Bitcoin 2026, a global open-source development program where contributors work on Bitcoin-related projects and infrastructure.
- Secured Rank 567 out of 1,94,752 candidates in AP EAPCET 2022, placing in the top 0.3% statewide.
- Achieved a score of 96.79% in the Joint Entrance Examination (IIT-JEE 2022) and received 10 medals for academic excellence during Class 11-12.

Leadership & Community

- **GSSoC Open-Source Contributor 2025** – GirlScript Summer of Code: Selected contributor making impactful contributions across open-source repositories, demonstrating collaborative engineering practices.
- **NSS Volunteer & Cyber Security Mentor – VRSEC:** Delivered a Cyber Security Awareness Program in a government school, translating technical knowledge into community impact.

Coding Profiles

- **LeetCode:** leetcode.com/SaiUmeshGupta
- **CodeChef:** codechef.com/users/saiumeshgupta5
- **HackerRank:** hackerrank.com/umeshgupta_05

Open Source Contributions

math-utils

Java, Maven

[io.github.umeshgupta05:math-utils:v1.0.0](https://github.com/umeshgupta05/math-utils) — Published an open-source Java 17+ utility library to Maven Central, exposing 12 APIs in 4 modules: prime operations, number theory, factorization and modular arithmetic; packaged with full input validation, overflow handling, and long variant support for large integer operations

DIPY (Diffusion Imaging in Python)

Python

[PR #3696](#) - Merged PR improving code correctness and test coverage in a widely used scientific computing library.